Engineering Decision-Making and Optimization

Hurwicz criterion under uncertainty, linear programming for a coal-fired plant, and weighted merit-function material selection

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Course	ME 403 Engineering Design, NJIT	**Software**	MATLAB (linprog, matrix operations), Microsoft Excel
Team	Samuel Cohen, Luis Ramirez, Larimel Rosario, Diego Arias, Miguel Huayahuas	**Inputs**	Problem data set from the last two and three digits of the student ID (…87, …287)

PROJECT SUMMARY

Three engineering decisions were solved with formal methods. First, the Hurwicz criterion was applied to choose between three vehicle programs under an uncertain economy; the SUV gave the highest value for every coefficient of optimism, and the lines only intersect outside the valid range ($\alpha = -1$ and -2). Second, the hourly burn rates of two coals were optimized with linear programming under emissions, conveyor, and pulverizer limits; the optimum of 12.21 tons/hr of coal A and 5.69 tons/hr of coal B produces 411,887 lb of steam per hour. Third, five metals were ranked for a robotic arm with a normalized weighted merit function, and 3.5% Ni, 0.4% C steel ranked first with M = 0.833.

1. Introduction

Design decisions often have to be made with incomplete information, competing limits, or several attributes that pull in different directions. This assignment applied three standard tools to these situations: a decision criterion for choosing under uncertainty, linear programming for allocating resources under constraints, and a weighted merit function for ranking alternatives across multiple attributes. Each problem was set up by hand and solved or checked in MATLAB.

2. Choice Under Uncertainty: Hurwicz Criterion

An automaker must choose between a sedan, crossover, or SUV program without knowing whether the economy will be strong or weak. The Hurwicz criterion blends the best and worst outcome of each choice using a coefficient of optimism α: $H = \alpha \cdot (\text{best}) + (1 - \alpha) \cdot (\text{worst})$. With $\alpha = 0$ the decision maker is fully pessimistic, and with $\alpha = 1$ fully optimistic.

Table 1. Rates of return and Hurwicz lines

Alternative	Strong economy	Weak economy	Hurwicz value H(α)
Sedan	28.7%	18.7%	18.7 + 10α
Crossover	38.7%	28.7%	28.7 + 10α
SUV	58.7%	38.7%	38.7 + 20α

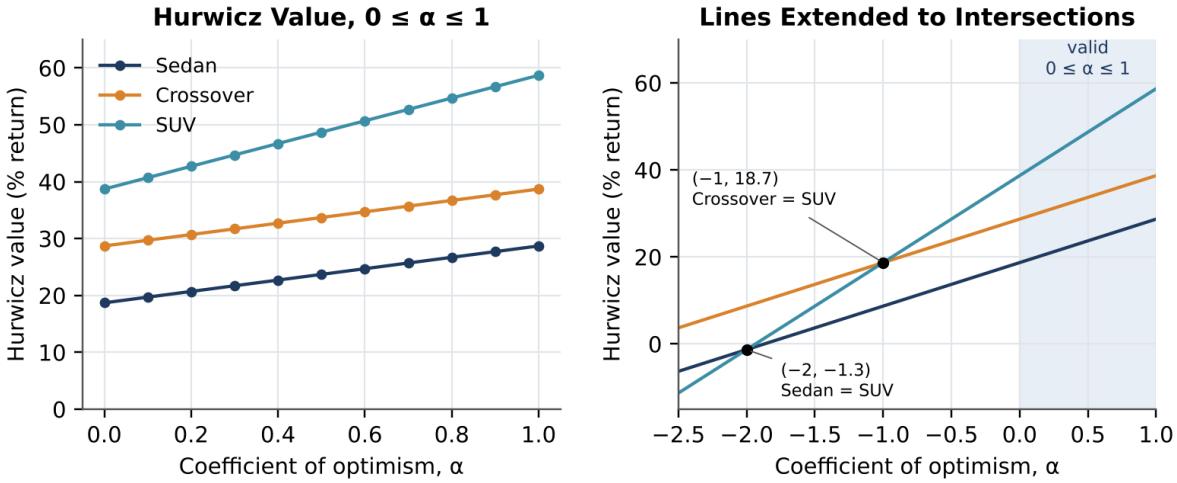


Figure 1. Hurwicz value vs. coefficient of optimism (left) and the lines extended to their intersections (right).

The SUV has the highest value at every α between 0 and 1, so it is the preferred choice regardless of the decision maker's optimism. The sedan and crossover lines are parallel and never meet, the SUV line crosses the crossover line at (α, H) = (−1, 18.7%) and the sedan line at (−2, −1.3%), and both intersections fall outside the valid range.

3. Linear Programming: Coal Burn Rates

A power plant burns coal A and coal B together and wants to maximize hourly steam production, which is directly tied to power output. The design variables are the burn rates x_1 and x_2 in tons/hr.

Table 2. Linear programming formulation

Item	Expression
Objective (maximize steam, lb/hr)	$f = 24{,}287x_1 + 20{,}287x_2$
Emissions limit (lb/hr)	$1.1287x_1 + 2.2287x_2 \le 26.4555$
Conveyor capacity (tons/hr)	$x_1 + x_2 \le 20$
Pulverizer time (hr per hr)	$x_1/16 + x_2/24 \le 1$ (equivalent to $3x_1 + 2x_2 \le 48$)
Bounds	$x_1, x_2 \ge 0$

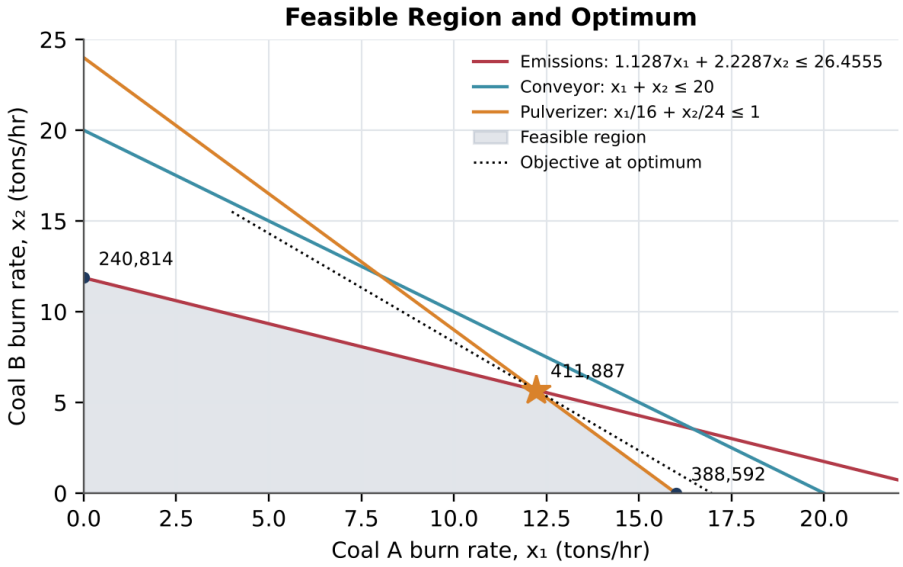


Figure 2. Feasible region, constraint lines, and steam output (lb/hr) at each corner point; the star marks the optimum.

MATLAB's *linprog* (maximizing by minimizing −f) returned x_1 = 12.21 tons/hr and x_2 = 5.69 tons/hr, giving a maximum of 411,887 lb of steam per hour. The graphical solution confirms this: the optimum sits at the corner where the emissions and pulverizer limits meet, both of which are fully used. The conveyor is not binding, since only 17.9 of its 20 tons/hr capacity is needed. Burning only coal A (16 tons/hr) would give 388,592 lb/hr, so mixing in the lower-emission coal B adds about 6% more steam within the same emissions limit.

4. Material Selection: Weighted Merit Function

Five metals were compared for a robotic arm in an ammonia plant. Each attribute was normalized from 0 (worst) to 1 (best): $R = (x - x_{min})/(x_{max} - x_{min})$ for yield strength, stiffness, and corrosion resistance, where higher is better, and $R = (x_{max} - x)/(x_{max} - x_{min})$ for density and cost, where lower is better. Importance weights came from rank points (E = 5, S_y = 4, corrosion = 3, cost = 2, density = 1) divided by their sum of 15, and the merit value is $M_i = \Sigma\, I_j R_{ij}$.

Table 3. Attribute data, normalized ratings R_{ij}, and merit values (weights in header)

Alternative	S_y (ksi) I = 0.267	E (10^6 psi) I = 0.333	Corrosion I = 0.200	Density (lbm/ft³) I = 0.067	Cost ($/ft) I = 0.133	M_i	Rank
Malleable cast iron	54 (0.148)	25 (0.742)	1 (0.000)	457.5 (0.092)	22 (0.970)	0.4223	4
Grey cast iron	30 (0.008)	13 (0.124)	1 (0.000)	445.8 (0.124)	20 (1.000)	0.1849	5
3.5% Ni, 0.4% C steel	200 (1.000)	30 (1.000)	2 (0.500)	490.8 (0.000)	20 (1.000)	0.8333	1
Al alloy 2024-T4	60 (0.183)	10.6 (0.000)	3 (1.000)	172.8 (0.878)	27 (0.896)	0.4267	3
Metal X	28.7 (0.000)	28.7 (0.933)	3 (1.000)	128.7 (1.000)	87 (0.000)	0.5777	2

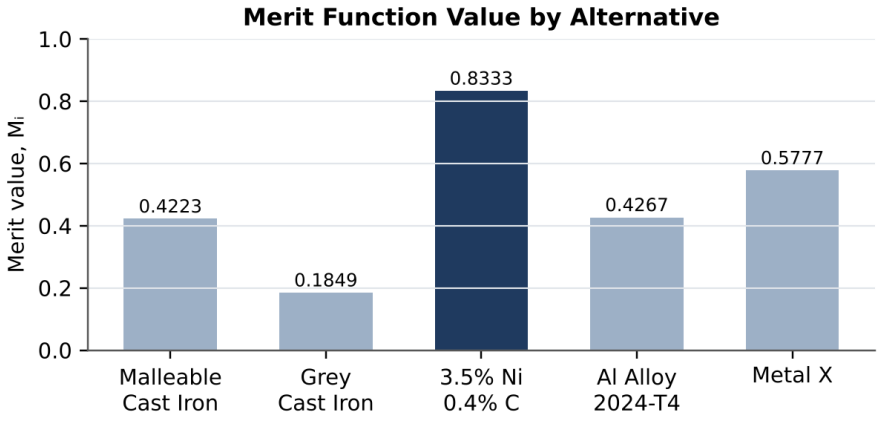


Figure 3. Merit function value for each candidate material.

The 3.5% Ni, 0.4% C steel ranked first (M = 0.833) because it has the highest yield strength and stiffness, the two most heavily weighted attributes, at the lowest cost. Metal X placed second on its low density, good stiffness, and corrosion resistance, but its high cost and lowest strength held it back. Aluminum 2024-T4 and malleable cast iron were nearly tied for third and fourth, and grey cast iron ranked last with low strength and stiffness.

5. Conclusion

The Hurwicz analysis showed the SUV program is preferred for any level of optimism, the linear program found an optimal mix of 12.21 tons/hr of coal A and 5.69 tons/hr of coal B for 411,887 lb/hr of steam with the emissions and pulverizer limits binding, and the merit function selected 3.5% Ni, 0.4% C steel for the robotic arm. Writing each problem in MATLAB made the results easy to check and to rerun if the inputs or weights change.

References

New Jersey Institute of Technology. (2026). *ME 403 Assignment 2: Optimization problems*. Department of Mechanical and Industrial Engineering.

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The MathWorks, Inc. (2026). *linprog: Solve linear programming problems* [Documentation].